International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Mid-Term Examination, Spring-2023

Course Code: **PHY 1201** Course Title: **Physics-II**

Time: 1 hour 30 minutes Full Marks: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs) of the Questions	
CO1	**Understand** the basic knowledge of different areas of physics like electromagnetism, structure of matter and modern physics as well as engineering aspect.
CO2	Apply mathematical knowledge to formulate and solve engineering problems.

Bloom's Levels of the Questions						
Letter Symbols	**R**	**U**	**App**	**An**	**E**	**C**
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

Part A

[Answer the questions from the followings]

a)	State and explain Coulomb's law for electrostatics.	CO1	R	2
b)	Define electric dipole. Derive an expression for electric potential and electric field due to an electric dipole.	CO1	U	5
c)	Calculate the electric potential and electric field due to a short dipole of dipole moment 2×10^{-27} coul-meter at a point distance 2cm from it (i) on its axis. (ii) On its perpendicular bisector.	CO2	A	3

a)	Define magnetic field.	CO1	R	1
b)	State Biot-Savart law. Applying Biot-Savart law, derive an expression for the magnetic field due to a long straight wire carrying current.	CO1	U	6
	Or			
	Define electric flux. State and explain Gauss's law of electricity.			
c)	What is the magnetic field 10 cm away from a long wire carrying a current of 10 ampere?	CO2	A	3

a)	Define clearly with diagram what do you mean by alternating current.	CO1	R	
b)	Find out an expression for the growth of charge in capacitor and the decay of current in the circuit.	CO1	U	5
	Or			
	Define electric field and electric potential. Find an expression for electric field due to a point charge.			
c)	A 2μF capacitor is connected to 440 V DC source through a resistor of 0.25 mega ohm. Find the charge on the capacitor at 1 second.	CO2	A	3

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Mid-Term Exam, Autumn 2022

Course Code: **PHY 1201** Course Title: **Physics-II**

Time: 1 hour 30 minutes Full Marks: 30

(i) The figures in the right-hand margin indicate full marks

(ii) Course Outcomes and Bloom's Levels are mentioned in additional Columns

Course Outcomes (COs) of the Questions	
CO1	Understand the basic knowledge of different areas of physics like electromagnetism, structure of matter and modern physics as well as engineering aspect.
CO2	Apply mathematical knowledge to formulate and solve engineering problems.

Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

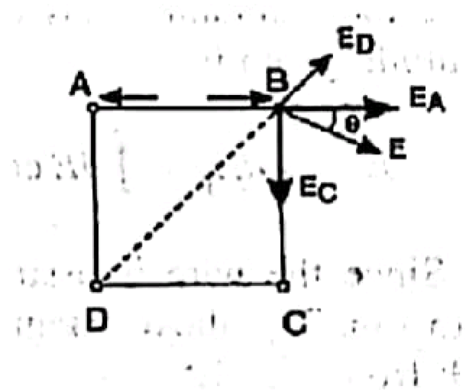
Part A

[Answer the questions from the followings]

1.	a)	State and explain Coulomb's law for electrostatics.	CO1	U	2
	b)	Derive an expression for an electric field due to a long uniformly charged wire. Or State and Explain Gauss's law for electrostatics.	CO1	U	5
	c)	Show that the electrical force between the electron and proton in the hydrogen atom is approximately 2.2×10^{39} times stronger than the gravitational forces.	CO2	A	3
2.	a)	State the Faraday's laws of electromagnetic induction.	CO1	U	2
	b)	Define self Inductance. Deduce a mathematical expression for the self inductance of a solenoid. Or Derive an expression for magnetic field at a point due to a long straight wire carrying current.	CO1	U	5
	c)	A solenoid having an air core and 100 cm long has 140 turns and its area of cross-section is 4 cm^2. Find the co-efficient of self-inductance of the solenoid.	CO2	A	3
3.	a)	What is RC Circuit?	CO1	R	1
	b)	Obtain an expression for the growth charge and current when a capacitor is charged through a resistance for a constant emf.	CO1	U	6
	c)	A capacitor of capacitance 0.1 μF is first charged and then discharged through a resistance of 10 megaohm. Find the time, the potential will take to fall to half its original value.	CO2	A	3

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE, Mid-Term Examination(Assignment), Spring-2022
Course Code: PHY- 1201 **Course Title: Physics-II**
Total marks: *21* Time: *3* hours

Answer All the Question

1. a)	Define electric dipole.	01
b)	Derive an expression for the electric field at a point on the equatorial line due to an electric dipole.	03
c)	ABCD is a square of **X** cm slide. Charges of 16×10^{-X}, -16×10^{-X} and 16×10^{-X} are placed at a point A, C and D respectively. Find the intensity of the electric field at a point B. **X= last digit of your ID , in case of 0, consider 3].**  Fig. 1(c)	03
2. a)	Define Magnetic Induction.	01
b)	Find the value of **B** due to a straight conductor carrying current loop of perpendicular distance x from the conductor.	03
c)	The magnetic field at a distance **X.Y** cm from the axis of a long straight wire is 13 mT. What is the current in the wire? **[X= last digit of your ID and Y= second last digit of your ID, in case of 0, consider 3].**	03
3. a)	What is time constant of RC Circuit?	01
b)	Show that the form factor is 1.11.	03
c)	An **X0** μF capacitor initially uncharged is connected through a X00 ohm resistor to a 12 V battery. **[X= last digit of your ID, in case of 0, consider 3].** (a) What is the magnitude of the final charge q_o on the capacitor? (b) What is time constant of the circuit? (c) How long after the capacitor is connected to the battery will it be charged to $\frac{1}{2}q_o$? (d) How long after the capacitor is connected to the battery will it be charged to $0.90q_o$?	03

Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Mid-term Examination, Autumn- 2018

Course Code: PHY- 1201 Course Title: Physics-II

Total marks: *30* Time: *1.5* hours

[Answer any *three* questions]

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1. a) Define electric field. 01
   b) Find the electric field at a point on the axis of a charged circular turn of wire. 06
   c) Show that the electrical force between the electron and proton in the hydrogen atom is approximately 2.2×10^{39} times stronger than the gravitational forces between them. 03

2. a) Define magnetic induction. 01
   b) State Biot-Savart law. Find the value of B due to a current loop of radius a carrying a current i at a distance x on the axis from the centre of the loop. What will be the value of B at the centre of the loop. 06
   c) A circular coil of radius 40 mm consists of 250 turns of wire in which the current is 20 mA. What is the magnetic induction in the centre of the coil? 03

3. a) Define RC Circuit. 01
   b) A capacitor is charged for a constant emf. and then allowed to discharge through a resistance. Find the expression for the discharge of the capacitor. 06
   c) A 1 μF capacitor is allowed to discharge through an unknown resistance. If the charge on the capacitor takes 31.65 seconds to drop to half of its original value, what is the value of the resistance? 03

4. a) Show that, the form factor for alternating current is 1.11. 01
   b) Derive an expression for the mean value of an alternating current. 03
   c) A square coil of side " *d* " carries a current " *I* ".Calculate the magnetic induction at the centre of the coil. 03
   d) Two coils, a primary of 600 turns and a secondary of 30 turns, are wound on an iron ring of mean radius 0.1 m and cross-section 2×10^{-2} m radius. Find their mutual inductance (μ_r for iron=600). 0[illegible]

Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Mid-term Examination, Spring- 2018

Course Code: PHY- 1201 Course Title: Physics-II

Total marks: *30* Time: *1.5* hours

[Answer any *three* questions]

a)	Define electric dipole.	01
b)	Find the expressions for the electric potential and electric field due to an electric dipole.	06
c)	A dipole is consisting of an electron and proton, 4×10^{-10} m apart. Compute the electric field at a distance of 2×10^{-8} m on a line making perpendicular with the dipole axis from the centre of the dipole.	03

a)	State the Faraday's laws of electromagnetic induction.	01
b)	Define coefficient of self-induction of a coil..Deduce a mathematical expression for the self inductance of a solenoid.	06
c)	Calculate the self-inductance of a solenoid having 800 turns and length 1 m. The area of cross-section is 6×10^{-4} m^2 and the relative permeability of the core is 900.	03

a)	Define R-C circuit.	01
b)	Obtain an expression for the growth of charge and current in a capacitor through a resistance.	06
c)	A 2μF capacitor is connected to 440 V DC source through a resistor of 0.25 mega ohm. Find the charge on the capacitor at 1 second.	03

a)	Define magnetic induction.	01
b)	State and explain Coulomb's law for electrostatics.	03
c)	Write down the magnetic properties of solid.	03
d)	Find the magnetic field at a perpendicular distance 30 cm from a straight wire carrying a current of 3 ampere.	03

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Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Mid-term Examination, Autumn- 2017

Course Code: PHY- 1201 Course Title: Physics-II

Total marks: *30* Time: *1.5* hours

[Answer any *three* questions]

Define electric field. 01

Find the electric field at a point on the axis of a charged circular turn of wire. 06

What is the magnitude of the electric field strength E such that an electron, placed in the field, would experience an electrical force equal to its weight? 03

Define magnetic induction. 01

Find the magnetic induction at any point on the axis of a solenoid. 06

A long solenoid of length 1 m and mean diameter 0.1 m consists of 1000 turns of wire. A current of 20 A flows through it. Calculate the magnetic induction on its axis (i) at its centre, (ii) at one of its ends. 03

, State the laws of Faraday and Lenz for electromagnetic induction. 02

, Define mutual inductance between a pair of coils. Deduce a mathematical expression for the mutual inductance between two coaxial solenoids. 05

, Two coils, a primary of 500 turns and a secondary of 30 turns, are wound on an iron ring of mean radius 0.1 m and cross-section 4×10^{-2} m diameter. Find their mutual inductance (μ_r for iron =800). 03

.) Define electromagnetic induction. 01

) State and explain Coulomb's law for electrostatics. 03

) Four equal point charges, +3mC, are placed at the four corners of a square that is 40 cm on a side. Find the force on any one of the charges. 03

) A circular coil of radius 40 mm consists of 250 turns of wire in which the current is 20 mA. What is the magnetic induction in the centre of the coil? 03

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Bismillahir Rahmanir Rahim

International Islamic University Chittagong

Department of Computer Science & Engineering

B. Sc. in CSE Mid-term Examination, Spring- 2016

Course Code: PHY- 1201 Course Title: Physics-II

Total marks: *30* Time: *1.5* hours

[Answer any *three* questions]

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1. a)	Define electric dipole.	01
b)	Find the expressions for the potential and field due to an electric dipole.	06
c)	Calculate the potential due to a short dipole of dipole moment 4.5×10^{-25} coul-metre at a distance of 2×10^{-8} m on a line making an angle 45° with the dipole axis from the centre of the dipole.	03
2. a)	Define magnetic induction.	01
b)	Find the magnetic induction at any point on the axis of a solenoid.	06
c)	A solenoid of 1200 turns is wound uniformly in a single layer on a glass tube 2m long and .2 m in diameter. Find the magnetic induction at the centre of the solenoid, when a current of 2A flows through it.	03
3. a)	State the Faraday's laws of electromagnetic induction.	02
b)	Deduce Faraday's law of electromagnetic induction in the form curl $E = -\frac{\partial B}{\partial t}$.	05
c)	A solenoid having a core of cross-section 4 cm^2, half air and half iron (relative permeability 500), is 22 cm long. If the number of turns on its 1000, what will be its self- inductance?	03
4. a)	Define electromagnetic induction.	01
b)	State and explain Biot-Savart's law.	03
c)	Two charges +q and -3q are separated by a distance of 1 m. At what points on its axis is the potential zero?	03
d)	Find the magnetic induction at the centre of a square current loop of side 1 meter carrying a current of 1 ampere.	03

8 7 E

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B. Sc. in CSE Mid-term Examination, Autumn- 2016

Course Code: PHY- 1201 Course Title: Physics-II

Total marks: *30* Time: *1.5* hours

15

[Answer any *three* questions]

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1. a) Define electric potential. 01
   b) Derive an expression for electric potential at a distance r from a point charge q. 06
   c) Find the potential at the centre of a 2.0 m square having charges q, -3q, 4q and 3q at its corners. ($q=1.0\times10^{-9}$ C). 03

2. a) Define magnetic induction. 01
   b) Using Biot-Savart law, calculate magnetic induction due to a current in a circular coil of wire at a point on its axis. 06
   c) A circular coil has a radius of 0.15 m and number of turns of 69. Calculate the magnetic induction at a point on the axis of the coil and distance .25m from the centre, when a current of 0.10 A flows in it. 03

3. a) State the Faraday's laws of electromagnetic induction. 02
   b) Define Mutual Inductance. Deduce a mathematical expression for the Mutual Inductance between two Coaxial Solenoids. 05
   c) Two coils, a primary of 800 turns and a secondary of 40 turns, are wound on an iron ring of mean radius 0.1 m and cross-section 2×10^{-2} m radius. Find their mutual inductance (μ_r for iron=800). 03

4. a) Define the terms: (i) Magnetic Flux (ii) Lorentz force 02
   b) State and explain Coulomb's law for electrostatics. 03
   c) Calculate the potential and field due to a short dipole of dipole moment 2×10^{-27} coul-meter at a point distance 2cm from it (i) on its axis. (ii) on its perpendicular bisector. 05